

Mario Makhlouta

mariomakhlouta13@gmail.com • +961 70 086 618 • Lebanon

github.com/MarioMakhlouta • linkedin.com/in/mario-makhlouta • mmakhlou.com

Objective

Computer & Software Engineering student with a strong foundation in programming, embedded systems, and cybersecurity. Hands-on experience through internships in cybersecurity and robotics, plus academic and personal projects in AI, FPGA design, and full-stack development. Seeking opportunities to apply my skills, contribute to impactful projects, and grow further in software engineering and any related fields.

Education

Bachelor of Engineering (BE) in Computer Engineering — University of Balamand (UOB)

2026 – 2027

Faculty of Engineering. One-year BE track following the BS in Computer Engineering.

Bachelor of Science (BS) in Computer Engineering — University of Balamand (UOB)

2022 – Spring 2026

Relevant courses: Data Structures, AI for Engineers, Operating Systems, Telecommunications Lab, Microprocessors, Programmable Logic Controllers (PLC), Cybersecurity, Computer Embedded Systems, Computer Architecture, Electronics.

Selected academic projects: Pipelined Mini-RISC-V Processor (CPEN314), Parallel Sorter using Bitonic Sort in SystemVerilog on DE1-SoC FPGA (CPEN313), PLC Automated Conveyor System (CPEN307), Microwave Timer using PIC18F45K22 (CPEN305), Privilege Escalation in Linux/Unix systems (CPEN341), SSB-AM Communication System Simulation (ELEN341), Delivery System in Java/OOP (CSIS215), Arithmetic Logic Unit in Quartus Prime (CPEN202), Scientific Calculator in MATLAB (ELEN202), Secure Multi-threaded Banking System (CSIS221).

Software Engineering Student — 42 Beirut

2024 – September 2026

Peer-to-peer learning, project-based curriculum in C/C++. Projects: push_swap, FdF, minishell, cub3D, C++ modules, ft_transcendence.

Experience

Robotics Intern & Trainee — InMind Academy / InMind Company

Summer 2026

- Two-month combined training and internship program in the robotics track.

Cybersecurity Intern — Prodigy InfoTech (Remote)

April 2026

- Completed a one-month remote internship covering 5 hands-on cybersecurity tasks.

Technical Skills

Programming	C, C++, Python, Java, SQL, Bash, Assembly (RISC-V, PIC18), SystemVerilog, PLC Ladder Logic, MATLAB
AI & Data	AI for Engineers fundamentals, Data Structures & Algorithms, problem-solving
Systems & OS	Operating Systems concepts, Unix/Linux, Memory management, Processes & threads
Embedded / FPGA	Quartus Prime, MPLAB, Multisim, Proteus, Arduino, PIC18, DE1-SoC, NIOS-V, Pipelined Processor Design, CPU Pipeline Hazards
Cybersecurity	Penetration testing fundamentals, Privilege escalation (Linux/Unix), Network security basics, Ethical hacking practice
Telecommunications	SSB-AM simulation, Telecom Lab measurements, Signal analysis
Web & Software	Object-Oriented Programming, Backend (Python), Full-stack basics, Git, Visual Studio
Databases	SQL, MariaDB, MS Access
Tools	Git/GitHub, Simulink, AutoCAD, Microsoft Office
Languages	Arabic (native), English (intermediate), French (intermediate)

Extracurricular Activities

- **Member — Institute of Electrical and Electronics Engineers (IEEE), UOB**
- **Arduino-Robotics Training Program — Generation of Innovation Leaders (GIL)** final project: smart box with sensor-controlled access.

Certificates

Cybersecurity Internship Completion — Prodigy InfoTech	2026
Robotics Internship & Training — InMind Company / InMind Academy	2026
OCB-051 Cybersecurity Bootcamp — Semicolon Academy	2025
IEEE Membership Certificate	2024
GDSC Board Membership Certificate — Google	2024
First Aid & CPR — Lebanese Red Cross	June 2024
42 Beirut Piscine Attendance Certificate	April 2024
MATLAB Onramp — MathWorks	March 2023
Robotics Training Program — GIL	2022
Full Stack Development Track — 1 million Arab Coders Initiative	November 2019